



SAPIENZA
UNIVERSITÀ DI ROMA

MASTER DEGREE IN ENGINEERING IN COMPUTER SCIENCE
& ARTIFICIAL INTELLIGENCE

DM – Project Proposal

Group Members

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Project Type

NoSQL – Graph Database vs Relational DBMS comparison

Dataset

Unified Host and Network Dataset (<https://csr.lanl.gov/data/2017/>)

Description

The goal of this project is to compare the performance of a relational database (PostgreSQL) with the one of a graph database (MemGraph), in order to see how they manage highly interconnected data. We decided to use MemGraph because we'd like to challenge ourselves and learn a new tool that was not covered during the Data Management course. We will use the "Unified Host and Network Dataset". In MemGraph, Users and Computers will be nodes, and network connections will be edges; in PostgreSQL we will create multiple separate tables for Users, Computers and Network Events, connecting them using foreign keys. We plan to run different queries on both systems, such as finding the shortest path a user can take to jump from one computer to another, and we will compare their execution times to analyze the pros and the cons of both technologies.